Efficient ML-Based Transient Thermal Prediction for 3D-ICs

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ABSTRACT

Thermal issues of 3D-ICs have become increasingly severe in recent years. Thus, thermal simulation is needed to ensure thermal safety during the design stage. However, performing thermal simulation iteratively requires a significant amount of time. As a result, a fast and accurate method for thermal prediction is a promising alternative to improve the turnaround time. In this paper, we propose a fast thermal prediction method using machine learning models. In the training phase, we employ two models: one for the initial three time steps and another for the subsequent time steps. To enhance prediction accuracy, we introduce two types of features: spaced-windowed features and time-decayed features. These features help us to capture spatial and temporal information effectively. In our experiment, the mean absolute error for the predicted temperature is 1.12°C, and the maximum error is 7.27 °C. In the prediction phase, we achieve a 116X speed-up compared to a commercial tool. With our proposed method, users can predict transient thermal profiles quickly and accurately to ensure thermal safety.

KEYWORDS

3D-IC, machine learning, thermal prediction

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1 INTRODUCTION

As semiconductor technology advances, the power density of designs continues to increase dramatically, leading to significant thermal issues [15]. In a 3D design, multiple layers are stacked vertically to reduce the total area and latency. However, this configuration concentrates heat sources since heat dissipation from the heated layer to the heat sink is inefficient [16]. Moreover, multiple active layers in a 3D design increase the total heat amount, further raising the temperature. Figure 1 shows the thermal profile of different dies in a 3D design, where die1 and die2 are both active layers. We observe that the temperatures of the two dies behave differently, with die2 exhibiting higher temperatures than die1. High temperatures result in large IR drops and longer RC delays, which degrade the performance and reliability of a chip [16]. Since performance and reliability are critical factors for automotive and artificial intelligence (AI) designs [12] [13], addressing thermal issues is essential.

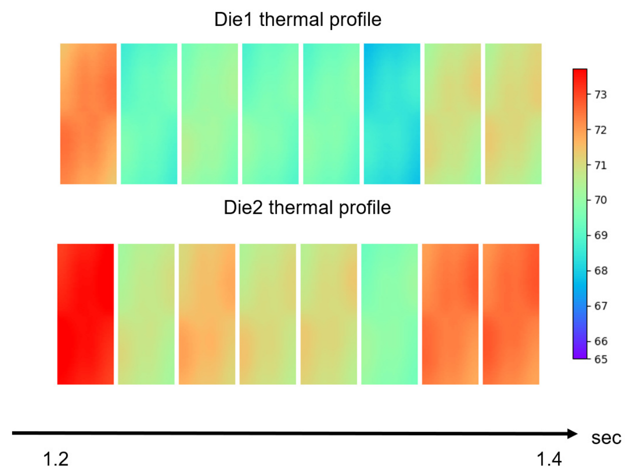

Figure 1: Thermal profile of different dies in a 3D design

Steady-state analysis generates a static thermal result for a given power map, while transient analysis generates a transient thermal profile for a given transient power profile over some time. Most commercial tools use the *finite-element method* (FEM) for 3D chip thermal analysis. This technique computes the numerical solution to a set of heat transfer equations [16]. FEM can be used for both steady-state and transient analysis. Although FEM can offer high accuracy, it requires a significant amount of computational resources. Additionally, thermal analysis is often iteratively performed during the design stage, leading to substantial runtime overhead. Therefore, a fast and accurate thermal prediction method is crucial for reducing this overhead and improving turnaround time. So far, there is no fast transient thermal predictor for 3D-ICs.

In this paper, we propose a transient thermal prediction method for 3D-ICs using machine learning. In the training phase, we first perform transient thermal simulations to obtain ground truth for our models. Then, we extract raw features from the design, power maps, and thermal profiles. After that, we create two types of features, *space-windowed features* and *time-decayed features.* These two types of features provide spatial and temporal information

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for each tile. Finally, we train two models: one for the initial three time steps and another for the subsequent time steps. Once both models are trained, we proceed to the prediction phase, where we can generate transient thermal profiles using our models without the need for further transient thermal simulations.

We conduct experiments on a 7nm industrial 3D design, considering three different power scenarios: only die1 is active, only die2 is active, and both dies are active. In our experiment, our proposed method achieves a mean absolute error (MAE) of 1.12°C and a maximum error (MaxE) of 7.27°C for predicted temperatures. The speedup ratio is 116X when comparing our prediction phase to a commercial tool. Thus, our proposed method can predict transient thermal profiles accurately in a short time.

The rest of the paper is organized as follows: Section 2 introduces previous works on thermal prediction, Section 3 explains the details of our proposed technique, Section 4 presents the experimental results, and Section 5 concludes the paper.

2 BACKGROUND

Several previous works are trying to speed up the process of thermal simulation. Juan *et al.* used auto-regressive Lasso model to predict the transient temperature of chip-multiprocessors [7]. Chhabria *et al.* proposed *ThermEDGe*, an encoder-decoder network, as a static and transient full-chip temperature predictor [3]. They used a U-Net-based model for static prediction and a Long short term memory (LSTM) based model for transient prediction. Ranade *et al.* introduced a machine learning (ML) based thermal solver, which can predict the temperature accurately on unseen cases with different chip sizes [14]. These works can predict the temperature quickly and precisely. However, they mainly focus on 2D designs.

Jin *et al.* proposed *ThermaGAN* to estimate the transient thermal profiles for commercial multi-core design based on the generative adversarial learning method [6]. They used the data from the thermal sensor to predict the whole thermal profile in real time. Nevertheless, this method requires post-silicon data during training. Liang *et al.* trained a convolutional neural network (CNN) to predict input power, then used a multi-scale decay surface model to predict transient thermal profiles [10] [9]. This method can predict the transient thermal profiles from ATPG test patterns. However, it also focuses on 2D designs only.

Kashyap *et al.* used conditional generative adversarial network (cGAN) to predict the thermal results of a 3D design [8]. The method can predict the thermal results of different layers when the layer is included in the training data. However, this method can only be used for static thermal prediction. Liu *et al.* introduced a framework *DeepOHeat*, to perform fast thermal prediction for 3D designs [11]. They applied *DeepONet* to solve heat equations instead of predicting temperatures. The method can perform well in unseen cases with different configurations. Although these two works can predict the temperature in 3D designs, they can only obtain static thermal results. Since no works are available for both transient thermal prediction for 3D designs, we need to develop a method to predict the transient thermal of 3D designs.

3 PROPOSED TECHNIQUE

Our proposed ML-based transient thermal prediction flow, illustrated in Figure 2, can be divided into two phases: the training phase and the prediction phase. The goal is to train machine learning models in the training phase and use these models to predict transient thermal profiles by input power maps during the prediction phase.

For the training phase, given the design, boundary conditions, and power maps, we first perform transient thermal simulation using RedHawk-SC-Electrothermal [5] to obtain transient thermal profiles. These results will be used as ground truth data to train our machine learning model. Next, we extract raw features using power maps from our dataset and thermal profiles obtained from RedHawk-SC-ET (Section 3.1). Subsequently, we create two types of features: *space-windowed features* (Section 3.2) and *time-decayed features* (Section 3.3). Finally, two machine learning models one for the initial three time steps, and the other for the remaining time steps are trained using the ground truth and all the aforementioned features (Section 3.4).

Once both models are trained, we then move on to the prediction phase. In this phase, transient thermal simulation is no longer required. We can directly extract raw features from power maps. Given the initial temperature, we can get the temperature result of the first time step. Then for each time step, the space-windowed and time-decayed features are created in the same manner as in the training phase. Finally, using these features as inputs to the trained models, we can obtain transient thermal profiles as our final results.

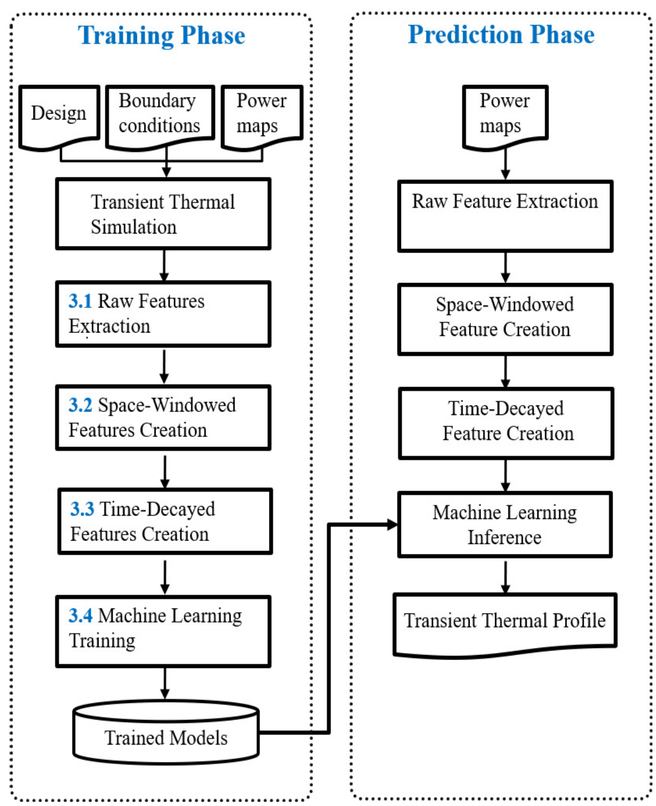

Figure 2: Proposed flow

Table 1: List of all raw features for a target tile i at time t

Category	Feature name	Source
Physical features	x coordinate	Design
	y coordinate	
	z coordinate	
Power features	die1 total power(t)	Power maps
	die2 total power(t)	
	combined total power(t)	
Thermal features	$tile_i$ temperature(t)	Thermal profiles
	avg. temperature(t)	

3.1 Raw Features Extraction

To train our models, we first need to extract raw features for every tile in each time step. According to *Joule's first law* (Eq. 1), the amount of heat Q that heats up the electrical conductor equals the product of the square of current I, resistance R and time t. Then by the equation of electrical power and *Ohm's law* (Eq. 2), we can substitute the square of the current multiplied by the resistance with power. Thus, we can conclude that the heat, which raises the temperature of the electrical conductor, is proportional to the power.

Once the heat is generated, it is then transferred to the heat sink via conduction. Through *Fourier's law of thermal conduction* (Eq. 3), we can see that the local heat flux density q equals the product of thermal conductivity k and the negative local temperature gradient $-\nabla T$. As a result, power and temperature are two key factors influencing thermal.

$$Q = I^2Rt \tag{1}$$

$$P = IV = I^2R \tag{2}$$

$$q = -k\nabla T \tag{3}$$

Additionally, since we aim to predict tile-based thermal profiles of different layers, physical information is also crucial. Therefore, we extract raw features and categorize them into three groups: *physical features, power features, and thermal features.* All raw features we extract are listed in Table 1.

Physical Features: Physical features include the x, y, and z coordinates of the tile. The x and y coordinates correspond to the center point of the tile, while the z coordinate is either 0 or 1 since we only consider the top and bottom die in our experiment.

Power Features: Given the tile-based power maps of both dies at time t, we calculate the total power of die1 (*(die1 total power(t)*) and die2 (*(die2 total power(t)*), respectively. We also sum up the power of both dies to get the combined total power (*combined total power(t)*).

Thermal Features: Transient thermal profiles we export from the simulation tool are converted into tile-based transient thermal profiles. For a target tile i at time t, we can use the temperature information at time t to predict the temperature at time t+Δt, where Δt is the size of one time step. $Tile_i$ *temperature(t)* indicates the temperature of target tile i in time step t. In addition to tile temperature, *avg. temperature(t)* is the average temperature among all tiles in time t, including top and bottom die.

3.2 Space-windowed Features Creation

It is insufficient to predict tile-based thermal profiles accurately if we only consider the total power and tile power since they provide too little information. Eq. 4 is a simplified 1D Fourier Law equation showing how heat is conducted in a 1D material [1], where Q is the amount of heat transferred in time step Δt, κ is the thermal conductivity, A is the area of the material, T_h is the temperature of the hotter end, T_c is the temperature of the colder end and d is the distance between hot and cold end.

$$\frac{Q}{\Delta t} = \frac{\kappa A(T_h - T_c)}{d} \tag{4}$$

As a result, for each target tile, raw features are not sufficient because the power of different locations may have varying impacts on the temperature. Thus, to obtain more precise information about power distribution than raw features for the machine learning model, we introduce spaced-windows features, which are inspired by [4].

We use three different window sizes: small, medium, and large to capture local, intermediate, and global effects, respectively. These windows help us to capture the impact of power at different scales. In our experiment, three window sizes are 80μm × 200μm, 160μm × 400μm, and 320μm × 800μm, respectively.

Three power maps, including power maps of die1, die2, and both dies are used to create space-windowed features in our experiment. Thus, nine space-windowed features will be created since we use three maps and select three window sizes in total. Thermal profiles are not used for space-windowed feature creation. The detailed implementation is illustrated below.

Figure 3 shows how we construct space-windowed features step by step, using the die1 power map as an example. First, we iterate through all the tiles, setting each as the *target tile* in turn (Figure 3 (a)). Then, for each target tile, we generate three windows of different sizes using the target tile as the center tile (Figure 3 (b)). Last, we calculate the average of all tile powers within each window to obtain the space-windowed features of the target tile (Figure 3 (c)).

3.3 Time-Decayed Features Creation

For each tile, the impact of power on temperature decays with distance and time by Equation 4 and [9]. Therefore, power in a previous time step can still affect thermal in subsequent time steps, which leads to the need for time-decayed features.

Before creating time-decayed features, we need to address two questions. The first question is how many backward time steps we need to consider. While considering more time steps provides more precise information, it also increases computation time and memory requirements. Conversely, considering too few time steps may result in the loss of important information. Thus, we need to strike a balance between information and computational resources. Our experiments indicate that the power of the last three time steps has the most significant impact on the subsequent temperature variation, so we choose to look at three time steps backward.

The second question is what kind of decay function to use. Based on our experiments, we found that exponential decay provides a good approximation, so we chose the exponential decay function. The formula of our function is shown in (Eq. 5), where $power_j(t -$

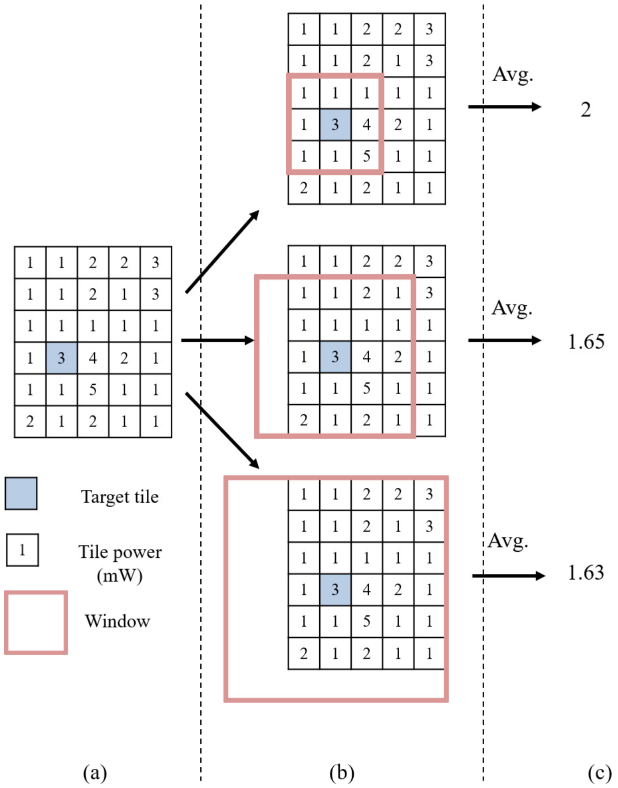

Figure 3: Illustration of space-windowed feature creation (a) select target tile, (b) generate three windows, (c) calculate the average of all tile values within each window

$k\Delta t$) indicates the power of tile j at time $t - \Delta tk$, t indicates the current time, k indicates the backward step index, and Δt is the size of one time step. α is the decay factor and it is set to $k+0.5$ in our experiment. Thus, the decay factor will become -1.5, -2.5, and -3.5 when k equals 1, 2, and 3, respectively.

$$\sum_{k=1}^{3} \left(power_j(t - k\Delta t) \times e^{-\alpha} \right) \tag{5}$$

We create twelve time-decayed features using all three raw power features and all nine space-windowed features. Finally, 29 features are used in total, including eight raw features, nine space-windowed features, and twelve time-decayed features. Once all the features are extracted and created, we can start to train the machine learning models.

3.4 Machine Learning Training

After all the features are ready, we can start to train the models. The model architecture we choose is XGBoost [2]. XGBoost, which stands for *eXtreme Gradient Boosting*, is a powerful machine learning algorithm widely used in both academia and industry. It is an implementation of *gradient boosted decision trees* designed for speed and performance.

Table 2: Model parameters

Parameter\Model	Model I	Model S
Max depth	5	7
Learning rate	0.03	0.03
Subsample rate	0.7	0.5
Colsample by tree	0.25	0.4
Sampling method	uniform	gradient based
Tree method	hist	hist
Objective	reg:squarederror	reg:squarederror

Since the temperature variation in the first three time steps is significantly different from the other time steps, we trained two different models: *Model I* and *Model S*. *Model I*, where *I* stands for initial, is designed to predict the delta temperature between the first three time steps, while *Model S*, where *S* stands for subsequent, predicts the remaining time steps. Therefore, we split the training data into two parts: one containing only the data of the first three time steps and the other containing the remaining time steps.

For both models, all the features mentioned above are used as input features. Given the input features of each tile at time t, the label of both models is the delta temperature between time t and $t+\Delta t$ of each tile, where Δt is the size of one time step. The number of boosting rounds is set to 1000 for both models and the parameters are listed in Table 2.

4 EXPERIMENTAL RESULTS

4.1 Experimental Setup

We conducted our experiments on a 7nm industrial 3D design. The 3D stackup of the design and the thickness of each component are shown in Figure 4. Die1 and Die2 are two identical dies without TSV, each measuring 0.32mm by 0.8mm. We generated various tile-based power maps for both dies, ranging from low to high power. Additionally, we considered three types of power scenarios: only die1 is active, only die2 is active, and both dies are active. The resolution of power maps is 50μm × 50μm.

We use RedHawk-SC-Electrothermal from Ansys to perform transient thermal simulation, setting the initial temperature and environment temperature to 25°C. Once the simulation is done, tile-based thermal profiles of the top and bottom die are extracted. The resolution of thermal profiles is 10μm × 10μm, and the duration of each time step is 10ms. We performed 17 transient thermal simulations in total, with each simulation covering 1.5 seconds. Twelve of these simulations were used as training cases and the remaining five were used as test cases.

Our experiments were executed using two different machines. For transient thermal simulations, we used a machine with an Intel Xeon Silver 4214 CPU and 128GB of system memory. For feature creation and machine learning training, we utilized a machine with an Intel Core i9-12900KF CPU, an Nvidia GeForce RTX 3090 GPU, and 128GB of system memory.

In our experiment, Mean Absolute Error (MAE), Max Error (MaxE), and Mean Error (ME) are chosen as our evaluation metrics.

$$\text{MAE} = \frac{1}{T \times N} \sum_{t} \sum_{i} |\hat{y}_i(t) - y_i(t)| \tag{6}$$

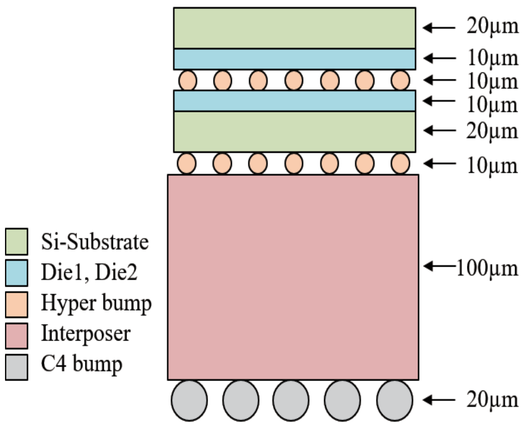

Figure 4: 3D stack up of the design and thickness of each component

$$\text{MaxE} = \max\left(\hat{y}_i(t) - y_i(t)\right) \quad \forall i, t \tag{7}$$

$$\text{ME} = \frac{1}{T \times N} \sum_t \sum_i \left(\hat{y}_i(t) - y_i(t)\right) \tag{8}$$

MAE (Eq. 6) indicates the average absolute prediction error, where T is the total number of time steps, N is the number of tiles, t stands for time step t, i stands for tile i, $\hat{y}_i(t)$ represents the predicted value of tile i at time step t, and $y_i(t)$ represents the ground truth of tile i at time step t. MaxE (Eq. 7) indicates the maximum prediction error among all time steps and tiles. If MaxE is greater than zero, it means that we over-predict in the worst case. Conversely, if MaxE is less than zero, it means that we under-predict in the worst case. ME (Eq. 8) shows the average prediction error across all time steps and tiles. If ME is close to zero, it means that the overestimated values and underestimated values offset each other over time.

4.2 Overall Results

Table 3 shows the overall results of the predicted temperature. For each test case, we calculate the MAE and MaxE for Die1, Die2, and both dies combined. In test case1 and test case2, only one die is active, while in the other cases, both dies are active. Our method demonstrates accurate temperature prediction, with the largest MAE being less than 2°C and the worst MaxE being less than 7.5°C. Table 4 shows the results between different feature sets. We compare the results among three different feature sets: raw features only, raw features + space-windowed features, and raw features + space-windowed features + time-decayed features. The MAE and MaxE of five test cases are displayed in the tables. After using our proposed features, the prediction results improve significantly, with the overall MAE decreasing from 1.63°C to 1.12°C and MaxE dropping from 11.43°C to -7.27°C. In conclusion, our proposed space-windowed features and time-decayed features enhance overall prediction performance.

Figure 5 and Figure 6 compare the golden thermal profile to the predicted thermal profile using one of the test cases. The golden thermal profile is generated using the commercial tool, while the predicted thermal profile is produced using our proposed method. The results of die1 are shown in Figure 5 and the results of die2 are shown in Figure 6, respectively. By comparing the results, we can observe that the predicted maps closely resemble the golden maps. Therefore, our method accurately predicts transient thermal profiles of 3D design.

Table 3: Overall results of predicted temperature

Map	Die1		Die2		All	
Metric (°C)	MAE	MaxE	MAE	MaxE	MAE	MaxE
Testcase1	0.83	4.10	0.77	4.47	0.80	4.47
Testcase2	0.68	3.23	0.69	-3.17	0.68	3.23
Testcase3	1.72	-6.47	1.96	-6.66	1.84	-6.66
Testcase4	1.13	-4.78	1.22	-4.91	1.17	-4.91
Testcase5	1.04	-6.78	1.17	-7.27	1.11	-7.27
	1.08	-6.78	1.16	-7.27	1.12	-7.27

*Simulation duration for each test case: 1.5 seconds

Table 4: Results between different feature sets

Feature set	Raw features		Raw features + SW		Raw features +SW +TD	
Metric (°C)	MAE	MaxE	MAE	MaxE	MAE	MaxE
Testcase1	1.79	11.43	1.34	9.87	**0.80**	**4.47**
Testcase2	1.13	-8.18	0.80	5.09	**0.68**	**3.23**
Testcase3	2.24	-9.35	2.56	-9.81	**1.84**	**-6.66**
Testcase4	1.71	-7.84	1.45	-8.47	**1.17**	**-4.91**
Testcase5	1.30	**-6.00**	1.46	7.02	**1.11**	-7.27
	1.63	11.43	1.52	9.81	**1.12**	**-7.27**

SW: space-windowed features TD: time-decayed features
*Simulation duration for each test case: 1.5 seconds

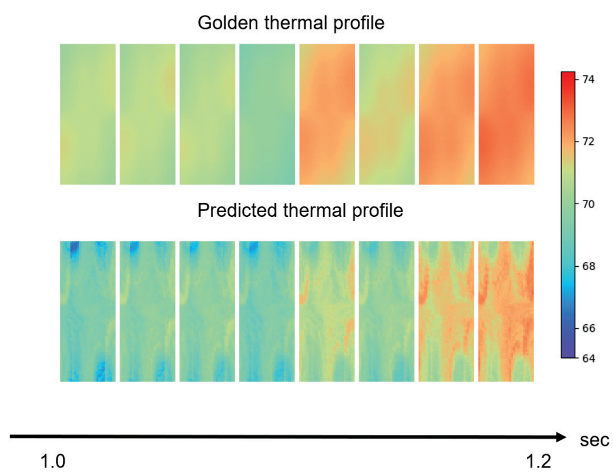

Figure 5: Predicted and golden thermal profile of die1

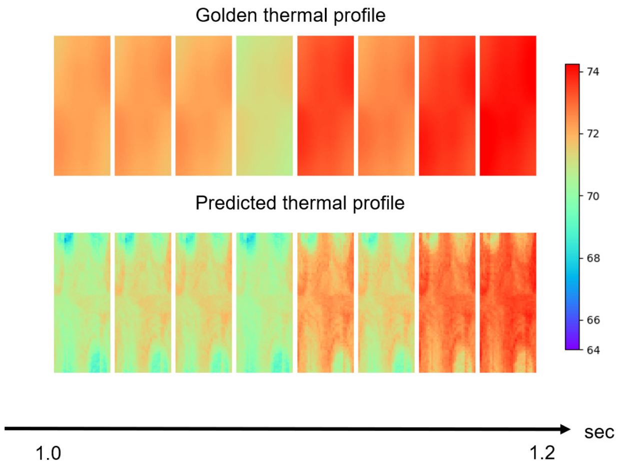

Figure 6: Predicted and golden thermal profile of die2

Table 5: Runtime of commercial tool, our training phase, and our prediction phase

	Comm-ercial	**Our training Phase**			**Our prediction Phase**	
Stage	Sim.	Data	Pre.	Train	Pre.	Inf.
Time (sec)	2,789	33,468	232	95	19	5
Total time	2,789			33,795		24
Speedup	**1X**		-			**116X**

Data: Training data generation Pre.: Feature extraction and feature creation
Inf.: Model inference
*Time for prediction phase and commercial tool: perform 1.5-second prediction/simulation

4.3 Runtime Comparison

We compared the runtime of our method with a commercial tool, as shown in Table 5. In our training phase, we first generate training data, then extract and create features, and finally train the models. In the prediction phase, we only need to extract and create features and then use the trained model for inference. For the commercial tool, only simulation is required. Comparing our prediction phase to the commercial tool, we achieve a 116 times speedup. Thus, our method can predict transient thermal profiles in a significantly shorter period.

4.4 Error Analysis

Since thermal simulations can last for tens of seconds, we want to determine if the prediction error accumulates or increases over time. Table 6 compares the MAE and ME of the predicted temperature. We observe that ME is much smaller than MAE, indicating that the overestimated values and underestimated values cancel out each other over time. Thus, there isn't a bias over time. Figure 7 shows the prediction error over time for one of the test cases, with the x-axis representing time and the y-axis representing the mean error. The blue line indicates the ME for each time step and the orange horizontal line represents the error equals zero. From the figure, we can see that the prediction error does not increase over time. As a result, although prediction errors are inevitable, they do not accumulate significantly over time during our 1.5-second experiment.

Table 6: Comparison between MAE and ME

Metric (°C)	**Die1 MAE**	**Die1 ME**	**Die2 MAE**	**Die2 ME**	**All MAE**	**All ME**
Testcase1	0.83	0.54	0.77	0.37	0.80	0.46
Testcase2	0.68	0.14	0.69	0.18	0.68	0.16
Testcase3	1.72	-0.59	1.96	-1.00	1.84	-0.80
Testcase4	1.13	-0.79	1.22	-0.92	1.17	-0.86
Testcase5	1.04	-0.38	1.17	-0.71	1.11	-0.55
	1.08	**-0.22**	1.16	**-0.42**	1.12	**-0.32**

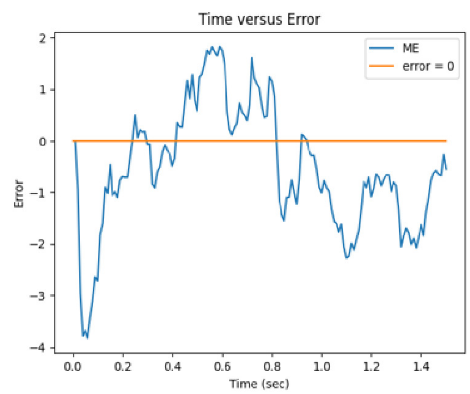

Figure 7: Prediction error over time

5 CONCLUSION

In this paper, we utilize machine learning to predict transient thermal profiles for 3D-ICs. We create space-windowed features to capture the spatial information of power maps, and we generate time-decayed features to obtain temporal information for power features. Both types of features effectively enhance model performance. We train two models: one for the initial three time steps and another for the subsequent time steps, as the behavior of the first three time steps differs from the others. In our experiments, we achieve a mean absolute error of 1.12°C and a maximum error of 7.27°C for the predicted temperature. Compared to a commercial tool, our method achieves a 116X speed-up. Using our proposed method, we can predict transient thermal profiles rapidly and precisely, thus ensuring thermal safety.

However, there are some limitations to our work. First, our model can not be transferred to other designs. Thus, we need to re-train new models if we want to predict the thermal profiles of a new design. Second, it takes a significant amount of time to generate training data during our training phase. Nevertheless, this can be amortized since we usually run the thermal simulations many times.

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